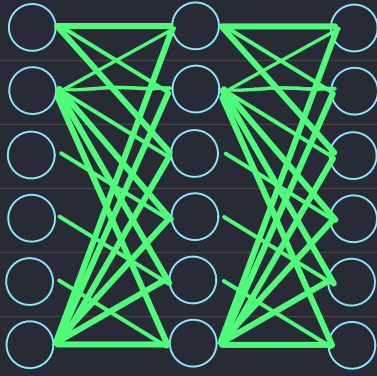


An Artificial Neuron - The Perceptron



A Neuron

上層的 n 個神經輸出



1. 權重與位移: $f = w_1 \times in_1 + w_2 \times in_2 + w_n \times in_n + b$
2. 套用激勵函式 (Activation Function): e.g. $y = \text{ReLU}(f)$

Activation Function

1. Tanh (Hyperbolic tangent)

$$f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

2. Sigmoid / Logistic

$$f(x) = \sigma(x) = \frac{1}{1 + e^{-x}}$$

3. ReLU (Rectified Linear Units)

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

4. Softplus

$$f(x) = \ln(1 + e^x)$$

